Rohan Singh

(650) 935-0809 | rohans9@uci.edu | linkedin.com/in/rohan123 | github.com/rohan1234usa | built-by-rohan.vercel.app

Education

University of California, Irvine Expected Grad: **June 2027**

B.S. in Computer Science, Specialization in Intelligent Systems ***(GPA: 3.92)*** *Dean's Honor List (All Quarters)*

- **Graduate study:** Applying to M.S. programs in AI & Computer Science for Fall 2027
- **Coursework:** ML & Data Mining, AI, Algorithms, Operating Systems, Databases, Information Retrieval
- **Publications:** 2 co-authored papers on facial emotion recognition – CVIP 2023 (Springer CCIS), WCSET 2023

Technical Skills

Languages: Python, Dart, TypeScript, JavaScript, C++, Java, Kotlin, SQL
AI/ML: PyTorch, TensorFlow, OpenCV, YOLOv8, DINOv2, Faiss, Gemini API, LangGraph, RAG, LLM grounding
Frameworks & Cloud: Flutter, Next.js, React, FastAPI, Firebase, PostgreSQL, Docker, AWS, GCP, GitHub Actions

Experience

Imentiv AI Cupertino, CA

Full-Stack Engineer Jun 2026 – Present

- Led PitchSense from brief to release: Imentiv's sales-call intelligence platform, with three products on one FastAPI + Next.js pipeline – Pitch Coach (the rep), Reaction (the audience), and Full Call (how one moves the other).
- Designed the scoring behind them: five engagement and six delivery constructs, each normalized to the speaker's own resting baseline so a reserved rep isn't graded on a showman's scale; the AI coach explains only those numbers.
- Built A/B testing for pitch variants (bootstrap 95% CIs, minimum-sample guardrails, ∼40 signals) and a harness metering every Imentiv call and credit, which cut ∼120 per-frame calls from each analysis.
- Designed SceneSense, a 0→1 acting coach that grades a take against the script's emotional arc; now building it.

AI/Machine Learning Intern Summer 2025

- Improved production YOLO face detection/recognition accuracy by 6% via automated ground-truth validation.
- Built face re-identification using dynamic placeholder tracks; validated against ByteTrack and BoT-FaceSORT.
- Benchmarked legacy vs. newer YOLO architectures; the results informed the team's decision to migrate.

AI/Machine Learning Intern Summer 2024

- Deduplicated training images with DINOv2 embeddings and Faiss search at a 95% similarity threshold.
- Built a YOLOv8 face-analysis tool and distinct-frame extraction to auto-label emotions in video datasets.

Merge Campus Irvine, CA

Founding Designer & Full-Stack Engineer Mar 2026 – Present

- **Product design:** Originated Merge Campus as a carpool app and widened it into a social platform for real-life plans with friends, defined through student personas, feature specs, and a 62-screen clickable prototype.
- **Architecture:** Lead system design as one of two engineers (750+ commits) for a Flutter + Firebase app with 110+ Cloud Functions – half its data model and an access model where the server decides who sees what.
- **Pipelines:** Built a Gemini place recommender that grounds and verifies its picks, receipt-to-split scanning, a timeline solver, and ride matching; raised expense-categorizer coverage from 39% to 76% at zero confident errors.
- **Security & quality:** Wrote most of the Firestore security rules and about three-quarters of the app's 1,600+ test files, and set up its first CI; fixes land test-first and are re-verified by deliberately breaking them.

Asterbyte Software Systems Remote

Software Engineer Intern Summer 2023

- Generated synthetic faces in Unreal Engine (MetaHumans) for EMOTE-2023, the dataset behind both papers.

Android Developer Intern Summer 2022

- Shipped Hue Christmas, a Kotlin Philips Hue light-show app (1K+ Google Play downloads); rebuilt Pitch Prime.

Projects

Behavioral Interview Coach | *Python, FastAPI, Next.js, Gemini, Imentiv API, AWS S3* | Live · Code

- Fuses face, voice, and transcript signals into four delivery scores: confidence, clarity, resilience, and engagement.
- Gemini streams résumé-tailored questions and turns past sessions into a personalized coaching plan.

SikhAI | *Next.js, TypeScript, Gemini, Firebase* | Live · Code

- Streaming Gemini chat through the lens of any of the ten Gurus, grounded in Gurbani fetched live (1,430 Angs).
- Schema-constrained JSON translator with a Cloud Translation fallback; nonce-fenced prompts against injection.